

SIMATS ENGINEERING

LAB EXPERIMENTS

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

COURSE CODE: CSA0515

EXP 2: DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, NOT NULL.

```
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.06 sec)

mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE Department ( Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50) NOT NULL UNIQUE);
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO Department VALUES^C
mysql> CREATE TABLE Department ( Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50) NOT NULL UNIQUE);
ERROR 1050 (42S01): Table 'department' already exists
mysql> INSERT INTO Department VALUES(101, 'Computer Science'),(102, 'Information Technology'),(103, 'Electronics');
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE Student (Student_ID INT PRIMARY KEY, Student_Name VARCHAR(50) NOT NULL, Email VARCHAR(100) UNIQUE, Age INT CHECK (Age >= 18), City VARCHAR(50) DEFAULT 'Chennai', Dept_ID INT, FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID));
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO Student(Student_ID, Student_Name, Email, Age, City, Dept_ID)VALUES(1, 'Dinesh', 'dinesh@gmail.com', 19, 'Chennai', 101),(2, 'Rahul', 'rahul@gmail.com', 20, 'Bangalore', 102),(3, 'Arun', 'arun@gmail.com', 18, DEFAULT, 103);
Query OK, 3 rows affected (0.06 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+-----+
| Student_ID | Student_Name | Email | Age | City | Dept_ID |
+-----+-----+-----+-----+-----+-----+
| 1 | Dinesh | dinesh@gmail.com | 19 | Chennai | 101 |
| 2 | Rahul | rahul@gmail.com | 20 | Bangalore | 102 |
| 3 | Arun | arun@gmail.com | 18 | Chennai | 103 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Student VALUES (1, 'Kiran', 'kiran@gmail.com', 20, 'Chennai', 101);
ERROR 1062 (23000): Duplicate entry '1' for key 'student.PRIMARY'
mysql> INSERT INTO Student VALUES (4, 'Kiran', 'dinesh@gmail.com', 20, 'Chennai', 101);
ERROR 1062 (23000): Duplicate entry 'dinesh@gmail.com' for key 'student.Email'
mysql> INSERT INTO Student VALUES (5, 'Ravi', 'ravi@gmail.com', 16, 'Chennai', 101);
ERROR 3819 (HY000): Check constraint 'student_chk_1' is violated.
mysql> INSERT INTO Student VALUES (6, 'Kumar', 'kumar@gmail.com', 20, 'Chennai', 999);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('collegedb`.`student`, CONSTRAINT `student_ibfk_1` FOREIGN KEY (`Dept_ID`) REFERENCES `department` (`Dept_ID`))
mysql> INSERT INTO Student VALUES (7, NULL, 'test@gmail.com', 20, 'Chennai', 101);
ERROR 1048 (23000): Column 'Student_Name' cannot be null
mysql> INSERT INTO Student (Student_ID, Student_Name, Email, Age, Dept_ID)VALUES(8, 'Vijay', 'vijay@gmail.com', 21, 101);
Query OK, 1 row affected (0.06 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+-----+
| Student_ID | Student_Name | Email | Age | City | Dept_ID |
+-----+-----+-----+-----+-----+-----+
| 1 | Dinesh | dinesh@gmail.com | 19 | Chennai | 101 |
| 2 | Rahul | rahul@gmail.com | 20 | Bangalore | 102 |
| 3 | Arun | arun@gmail.com | 18 | Chennai | 103 |
| 8 | Vijay | vijay@gmail.com | 21 | Chennai | 101 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```